



Customer Behavior Analysis

End-to-end data analyst project: Python → SQL → Power BI

Data Pipeline & Methodology

Ingest

Python / Pandas — read raw CSV

Clean

Standardize names, impute missing ratings

Feature Eng.

Age groups, purchase-frequency

Export

Clean CSV for SQL

Analyze

SQL Server — aggregations, window functions

Visualize

Power BI dashboard & PDF report

Python Cleaning Snapshot

Key steps condensed:

Standardize

Lowercase columns; replace spaces with underscores

Impute Ratings

Fill missing review_rating with category median

Age Groups

pd.qcut → younger_adult, adult, middle_age, senior

Cleanup

Drop promo_code_used; export cleaned CSV

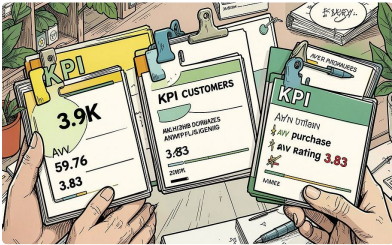


SQL Techniques

- GROUP BY, SUM / AVG / COUNT
- CTEs & window functions (ROW_NUMBER)
- Subqueries, CASE, conditional aggregation

Used to answer business questions and rank products

Power BI Dashboard



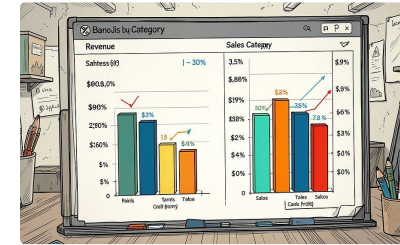
KPI Highlights

3,900 customers · Avg purchase 59.76 ·
Avg review 3.83



Interactive Filters

Slicers: subscription, gender, category



Category Visuals

Revenue & sales contribution by
category

Project Metrics

Records

3,900

Unique customers

3,900

Total revenue

233,081.00

Avg purchase

59.76

Avg rating

3.75

Representative Analyses

Selected business questions & outcomes

Q1 — Revenue by Gender

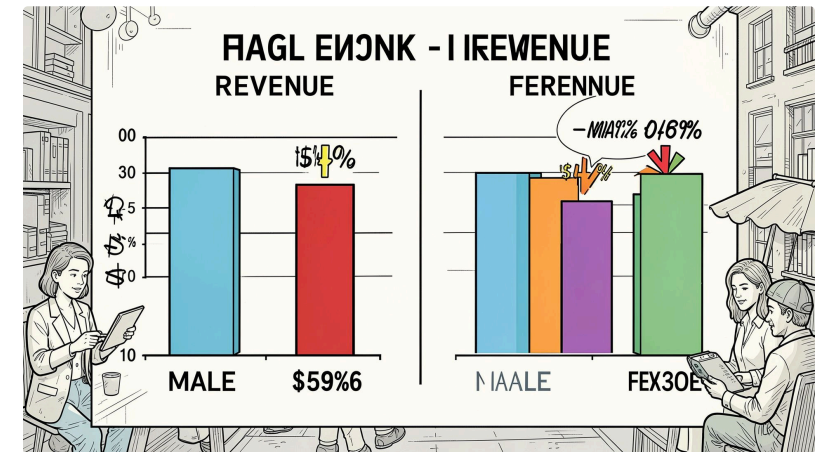
Male: 157,890 · Female: 75,191

Q4 — Shipping Avg

Express: 60.48 · Standard:
58.46

Q5 — Subscribers

No: avg 59.87 / revenue 170,436 · Yes: avg 59.49 / revenue 62,645



Product & Discount Insights



Top-rated products

Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat 3.80 · Skirt 3.78



High discount rates

Hat 50% · Sneakers 49.66% · Coat 49.07% · Sweater 48.17% ·
Pants 47.37%



Customer Segments & Behavior

Segments

New 83 · Returning 701 · Loyal 3,116

Repeat buyers & subscription

Repeat (>5): No 2,518 (72.44%) · Yes 958 (27.56%)

Top products by category

Accessories: jewelry, belt, sunglasses · Clothing: blouse, pants, shirt · Footwear: sneakers, shoes, slippers



Revenue by Age Group & Final Deliverable

62,143
younger_adult

59,197
middle_age

55,978
adult

55,763
senior

Final deliverables: cleaned dataset, SQL queries, Power BI dashboard, PDF report



Power BI dashboard: KPI cards, category charts, age-group visuals